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Specific Early Number Skills Mediate the Association

Between Executive Functioning Skills and Mathematics Achievement

Mary Wagner Fuhs[1], Caroline Byrd Hornburg[2], Nicole M. McNeil[2]

[1]Department of Psychology, University of Dayton

[2]Department of Psychology, University of Notre Dame

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Corresponding Author - Mary Wagner (Middle) Fuhs (Last): Department of Psychology, University of Dayton, SJ 319, 300 College Park, Dayton, OH 45469, mfuhs1@udayton.edu

Caroline Byrd (Middle) Hornburg (Last): Department of Psychology, University of Notre Dame, 118 Haggar Hall, Notre Dame, IN 46556, hornburg.1@nd.edu

Nicole M. McNeil: Department of Psychology, University of Notre Dame, 118 Haggar Hall, Notre Dame, IN 46556, nmcneil@nd.edu

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Keywords: mathematics achievement, number, executive functioning

Abstract

A growing literature reports significant associations between children's executive functioning skills and mathematics achievement. The purpose of this study was to examine if specific early number skills, such as quantity discrimination, number line estimation, number sets identification, fast counting, and number word comprehension mediate this association. In 141 kindergarteners, cross-sectional analyses controlling for IQ revealed that number sets identification (but not the other early number skills) mediated the association between executive functioning skills and mathematics achievement. A longitudinal analysis showed that higher executive functioning skills predicted higher number sets identification in kindergarten, which in turn predicted growth in mathematics achievement from kindergarten to second grade. Results suggest that executive functioning skills may help children quickly and accurately identify number sets as wholes instead of getting distracted by the individual components of the sets, and this focus on sets, in turn, may help children learn more advanced mathematics concepts in the early elementary grades.

Specific Early Number Skills Mediate the Association

Between Executive Functioning Skills and Mathematics Achievement

Mathematics achievement is one of the strongest predictors of later academic success in *both* reading and mathematics (Duncan et al., 2007). Thus, understanding patterns and predictors of mathematics achievement has become increasingly important not only for theoretical reasons, but also for the successful design of early mathematics curricula. One set of skills that is hypothesized to predict early mathematics achievement is executive functioning skills, including the ability to attend to relevant information and inhibit attention to distractors, hold information in mind and manipulate it, and shift attention. Many studies suggest a strong and persistent link between young children's executive functioning skills and their mathematics achievement (e.g., Blair & Razza, 2007; Bull & Lee, 2014; Bull, Espy, & Wiebe, 2008; Espy et al., 2004; Fuhs, Nesbitt, Farran, & Dong, 2014; Gilmore et al., 2013; Welsh, Nix, Blair, Bierman, & Nelson, 2010). However, the mechanisms involved in this link are not clear. In the current study, we tested the hypothesis that strong executive functioning skills help children construct specific early number skills that, in turn, facilitate mathematics achievement growth from kindergarten to second grade.

Executive Functioning Skills and Mathematics Achievement

Definitions of executive functioning skills typically include working memory (the ability to hold information in mind and manipulate it), inhibitory control skills (the ability to attend to relevant information and inhibit irrelevant or distracting information), and attention shifting or cognitive flexibility (the ability to shift flexibly between rule sets or task demands) (Miyake, Friedman, Emerson, Witzki, Howerter, & Wager, 2000). Recent empirical work assessing the dimensionality of this construct in young children (ages 3 – 5) has suggested that these skills may best reflect a one-factor construct in early childhood (e.g., Bull, Espy, Wiebe, Sheffield, & Nelson, 2011; Bull & Scerif, 2001; Hughes, Ensor, Wilson, & Graham, 2010; Wiebe, Espy, & Charak, 2008). A recent study of slightly older children (kindergarteners) has pointed to a possible two-factor model of executive functioning skills with inhibitory control and attention shifting loading on one factor and working memory loading on a separate factor (Lee, Bull, & Ho, 2013). Still, the general consensus is that there is tremendous overlap among executive functioning component skills in early to middle childhood, such that investigating executive functioning skills independently becomes quite difficult. This may reflect the lack of differentiation and specialization of the pre-frontal cortex at younger ages, but it also may reflect the difficulty in assessing only one subcomponent of executive functioning while not also assessing others (i.e., the task impurity problem). Furthermore, assessments of executive functioning skills also likely have considerable measurement error due to the fact that they also tap children’s language skills, motor skills, or other skills required for the particular task. Therefore, we use the term executive functioning skills in this paper while acknowledging that the assessments we use likely capture a combination of inhibitory control, working memory, and/or cognitive flexibility skills.

Executive functioning skills are linked to the functioning of the pre-frontal cortex and show an extended developmental trajectory into early adulthood (e.g., Anderson, 2002), with children exhibiting rapid changes during early childhood (e.g., Carlson, 2005; Garon, Bryson, & Smith, 2008). Significant individual variation in executive functioning skills and the extended developmental timescale of the pre-frontal cortex suggest that the environment may play a role in shaping the developmental trajectory of executive functioning skills in children. Several studies have found differences in executive functioning skills as a function of socio-economic status (e.g., Noble, Norman, & Farah, 2005), but other studies have found mixed results concerning whether or not improvements in executive functioning skills result from participating in interventions that either directly or indirectly target executive functioning skills (Farran & Wilson, under review; Raver et al., 2011; Tominey & McClelland, 2011). This recent attention to the malleability of executive functioning skills has been driven by research showing strong associations between executive functioning skills and positive outcomes in a variety of areas, including health and well-being (Moffitt et al., 2011) and academic achievement (e.g., McClelland et al., 2007).

Executive functioning skills are associated with better overall academic achievement in early childhood across content areas (e.g., Blair & Razza, 2007; Bull, Espy, & Wiebe, 2008; Clark et al., 2010; McClelland et al., 2007; Welsh et al., 2010). They may allow children to adapt to early learning environments generally by way of increasing children's attention to academic material in the face of distractions (e.g., Blair & Diamond, 2008). Specifically, executive functioning skills may work by increasing children's learning-related behaviors, which in turn increases academic achievement across content areas (Nesbitt, Farran, & Fuhs, 2015). For example, executive functioning skills may help children to remain engaged in academically-focused activities or refrain from disruptive behavior in a classroom. From this account, we might expect executive functioning skills to be related to a variety of academic content areas including both literacy and mathematics in early childhood. Indeed, support exists for a somewhat domain-general association between executive functioning skills and academic achievement in pre-kindergarten children (e.g., Allen & Lonigan, 2011; Fuhs, Farran, & Nesbitt, 2015; McClelland et al., 2007; Nesbitt et al., 2015).

Recent evidence, however, suggests that a specific and more direct link between executive functioning skills and mathematics achievement may emerge as early as kindergarten (Fuhs et al., 2014). Beyond pre-kindergarten, Blair, Knipe, and Gamson (2008) proposed that executive functioning skills may work to facilitate mathematics achievement in a more direct way than by facilitating more global learning-related behaviors in the classroom. Specifically, many of the skills on which mathematics achievement is built tend to require less rote processing and more online processing of information in the face of distractors, suggesting a continued role for executive functioning skills in mathematics achievement (e.g., Bull et al., 2008; De Smedt, Verschaffel, & Ghesquière, 2009; Raghubar, Barnes, & Hecht, 2010; St Clair-Thompson & Gathercole, 2006; Van der Ven, Kroesbergen, Boom, & Leseman, 2012). In contrast, the processing of letters and other early literacy skills, for example, becomes increasingly more automatic, and executive functioning skills may no longer play a significant role beyond early childhood.

At least two recent longitudinal studies support the idea that there is a unique association between executive functioning skills and mathematics achievement that does not also hold for literacy achievement. Brock, Rimm-Kaufman, Nathanson, and Grimm (2009) found that kindergarteners' executive functioning skills were associated with their mathematics achievement gains but not with their gains in literacy achievement. Fuhs et al. (2014) included assessments of children's executive functioning skills and academic achievement across three time points spanning from the beginning of pre-kindergarten to the end of kindergarten and found that children's executive functioning skills continued to predict gains in mathematics achievement and oral comprehension skills through kindergarten but did not predict gains in literacy achievement. Taken together, these studies suggest the possibility of a direct link between executive functioning skills and mathematics achievement that is robust through the transition into formal schooling.

In older children, different components of executive functioning skills may relate differently to mathematics achievement. The two specific components that have been most associated with mathematics achievement in older children are inhibitory control (e.g., Bull, Espy, & Wiebe, 2008; Espy, McDiarmid, Cwik, Stalets, Hamby, & Senn, 2004) and working memory or updating (Raghubar et al., 2010). Van der Ven and colleagues (2012) assessed children on a number of measures of updating, inhibitory control, and cognitive flexibility, and they found that when updating measures were included in their model, measures of inhibitory control and cognitive flexibility were no longer predictive of children's mathematics achievement. This finding could suggest that working memory skills are the most robust predictors of mathematics achievement, although Raghubar and colleagues (2010) noted that working memory tasks often include digit span tasks in which children and adults are asked to repeat a series of digits backwards. This standard practice can be problematic, as numerical information is part of the working memory task, leaving open questions about the uniqueness of that relationship outside of the common numerical information. Because of this and other considerations, including the mixed empirical support for differentiating components of executive functioning in young childhood, we conceptualize executive functioning skills as a single construct in the current study.

Executive Functioning Skills and Specific Early Number Skills

Prior research leaves unanswered questions about the possible mechanisms that explain *why* this association between executive functioning skills and mathematics achievement persists in childhood. Bull and Lee (2014) proposed that executive functioning skills could be related to mathematics achievement through a number of possible mechanisms. First, executive functioning skills may aid children in storing and manipulating relevant information during a mathematical problem. Executive functioning skills could also facilitate appropriate strategy use when inappropriate arithmetical strategies are more salient (e.g., using the “min” strategy–counting from the larger addend–instead of counting up from the first addend in an addition problem), attending to appropriate number representations (e.g., focusing on a fraction as a whole rather than on the individual numbers that make up the numerator and denominator), or attending to relevant information in a word problem (e.g., attending to the numerical information in a story problem about candies being divided instead of becoming distracted by salient memories or associations with candy). Finally, executive functioning skills may aid children in flexibly shifting between representations of number. Consistent with these predictions, Lubin, Vidal, Lanoe, Houde, and Borst (2013) found that executive functioning skills predicted sixth-grade students’ ability to respond correctly to word problems when the relational term (e.g., more than) was inconsistent with the arithmetical operation required to correctly solve the problem (e.g., subtraction). Siegler and Pyke (2013) found at least partial support for a link between executive functioning skills and fraction knowledge in middle-school students, suggesting that executive functioning skills helped children attend to the properties of the fraction as a whole, rather than becoming distracted by the symbolic quantity of the numerator or denominator alone. These studies suggest generally that executive functioning skills may facilitate mathematics success in older children on tasks that require flexibility in strategy use and/or attention to multiple aspects of a numerical problem as a whole rather than becoming distracted by specific components of the problem that may be salient.

These same processes may be at work in younger children who have primarily received informal exposure to early number concepts and skills prior to kindergarten. In fact, because kindergarten children are still in a period of rapid executive functioning skills development, it can be argued that they are perhaps *more* likely than older children to be distracted by specific salient components of a mathematics problem. This may manifest as difficulty attending to an Arabic numeral as a whole (e.g., 12) rather than on the individual numbers that make up the number (e.g., 1 and 2) or as difficulty attending to a number line as a whole rather than being pulled to one of the endpoints. This may also manifest as difficulty keeping track when counting objects or as difficulty attending to a set of symbolic or non-symbolic objects as a whole (total amount) instead of focusing on specific components of the set or features of the individual objects in the set.

Executive functioning skills may aid children in these types of specific early number skills, which should then promote fluency with and conceptual understanding of numbers, which should ultimately facilitate more formal mathematics skills such as symbolic arithmetic calculation and word problem solving. Note the distinction here between specific early number skills, which are developed more informally prior to the start of elementary school mathematics instruction, and formal mathematics achievement in school. The informal early number skills that children develop prior to formal instruction in mathematics are sometimes collectively referred to as “number sense” (Jordan, Kaplan, Oláh, & Locuniak, 2006). Although no two researchers use the term “number sense” in precisely the same way (Berch, 2005), many would agree that these early number skills draw on a mental number line and include behaviors such as counting, estimating and comparing numerical quantities, identifying important numerical relationships, and performing simple number transformations (Dehaene, 1997; Griffin, Case, &

Siegler, 1994; Jordan et al., 2006; National Council of Teachers of Mathematics, 2000). Such skills provide the foundation for later formal mathematics learning in school (Aunola, Leskinen, Lerkkanen, & Nurmi, 2004; Geary, Hoard, Nugent, & Bailey, 2013; Jordan et al., 2012; Jordan, Kaplan, Locuniak, & Ramineni, 2007; Hornung, Schlitz, Brunner, & Martin, 2014).

Unfortunately, there are fewer prospective studies in the literature pointing to specific mechanisms to explain the link between executive functioning skills and mathematics achievement in young children, but recent research does point to associations among executive functioning skills and some of the specific early number skills mentioned above. For example, some evidence supports an association between the central executive components of working memory and symbolic quantity discrimination (Cirino, 2011; Krajewski & Schneider, 2009) and number line estimation skills in kindergarteners (Geary, Hoard, Nugent, & Byrd-Craven, 2008). Studies have also linked executive functioning skills to counting skills (Geary, Hoard, Byrd-Craven, Nugent, & Numtee, 2007; Kroesbergen, Van Luit, Van Lieshout, Van Loosbroek, & Van de Rijt, 2009) and to the ability to identify and combine sets consisting of non-symbolic and symbolic quantities (Geary et al., 2007).

Each of these specific early number skills, in turn, has been shown to predict mathematics achievement. For example, Lembke and Foegen (2009) found that quantity discrimination skills significantly predicted both kindergarten and first grade children's mathematics achievement, as assessed by both teacher ratings and standardized assessments. Counting skills have also been linked to mathematics achievement in recent longitudinal research (Bartelet, Vaessen, Blomert, & Ansari, 2014; Krajewski & Schneider, 2009). Finally, both number line estimation skills (e.g., Sasanguie, De Smedt, Defever, & Reynvoet, 2012; Siegler & Booth, 2004) and the ability to identify and combine sets of non-symbolic and symbolic quantities (Geary, Bailey, & Hoard, 2009) have both been shown to predict mathematics achievement.

In sum, a large body of evidence suggests that young children's executive functioning skills predict their general mathematics achievement. A smaller body of work suggests that executive functioning skills may predict more specific early number skills that are known to be important for future mathematics achievement. What is missing in prior work is an account to explain how these associations operate together. In the current study, we propose that specific early number skills mediate the association between children's executive functioning skills and their growth in mathematics achievement across time.

Current Study

We first assessed children on a battery of executive functioning skills tasks, early number skills tasks, and a standardized mathematics achievement measure during their kindergarten year. The specific early number skills we examined have been shown in prior research to be correlated with both executive functioning skills and mathematics achievement. Our goal was to determine if these early number skills mediate the well-established association between executive functioning skills and mathematics achievement. We also were interested in extending this study to capture mathematics achievement *gains* to test if the cross-sectional findings held longitudinally, so we assessed children's mathematics achievement again two years later in second grade. We predicted that executive functioning skills would facilitate specific early number skills, which in turn would predict mathematics achievement.

Method

Participants

We worked with 141 children (Time 1 *M* age = 6 yrs, 2 mo; 88 boys, 53 girls) from a mid-sized city in the Midwestern United States. The race/ethnicity of the sample was 7% African American or black, 3% Asian, 8% Hispanic or Latino, 11% other, and 71% white. Approximately 42% of children received free or reduced price lunch.

Measures

All assessments were videotaped and inter-rater reliability was established by having a second coder evaluate the responses of a randomly selected 20% subsample.

Executive Functioning Skills. *Day/Night Stroop Task (Gerstadt, Hong, & Diamond, 1994).* This measure has typically been conceptualized as an assessment of inhibitory control, although it also requires working memory, as children have to hold the rule in mind while inhibiting a salient but incorrect response. Children were told to play a "game of opposites," by saying "night" when shown a white card with a picture of the sun and "day" when shown a black card with a picture of the moon. There were 16 trials. Children were asked to respond as quickly as possible. Agreement between coders to test for inter-rater reliability was 99% for coding whether or not a given trial was correct. We used total number of correct responses as the outcome variable in analyses.

Dimensional Change Card Sort Task (DCCS; Zelazo, 2006). This task is typically conceptualized as primarily a task of cognitive flexibility or attention shifting, although inhibitory control is likely also required to complete this task successfully. Children were asked to play a card game under multiple sets of rules. Children were first asked to play the "shape game" by sorting cards based on their shape. After sorting cards by shape, children were told that the rules "changed" and that they should now play the "color game" and sort cards based on color. Children who were able to switch between the two games (able to successfully sort at least five of six cards after the rule switched) advanced to the "border version," in which half of the cards had a black border. In this final game, children were instructed to sort cards with the black border according to the "color game" and those without a border according to the "shape game." Inter-rater reliability was high; agreement between coders was 100% for coding total correct during the pre-switch (6 trials), post-switch (6 trials), and border version (12 trials) phases. We used total number of correct responses from the post-switch trials (out of 18) as the outcome variable in analyses.

Early Number Skills. *Quantity Discrimination (Lembke & Foegen, 2009).* Children were shown sets of number pairs ranging from 0-20 and were asked to mark the larger number in each pair (e.g., if the number pair was 12 and 17, children should mark 17). Children completed two sets and were given 60 seconds to mark as many as possible. Instructions were not repeated before the child started the second set. Inter-rater reliability was high; agreement between coders was 100% for coding the total number correct. We used total number of correct responses as the outcome variable in analyses.

Number Line Estimation (Siegler & Booth, 2004; Siegler & Opfer, 2003; see also Barth & Paladino, 2011). Children were given a line with endpoints 0 and 20. First, children were asked to place the number 10 on the number line. On the next page, the experimenter showed the child where 10 should go and indicated that 10 is the only number that goes halfway between 0 and 20, in order to help ensure that children understood the task. Then, children were asked to place a random number between 1 and 19 on the number line. They repeated this process on separate number lines until they had placed all values between 1 and 19. The accuracy of children's number line estimates were determined by calculating each child's proportion absolute error, which is a measure of the discrepancy between the child's marked values and the actual values over all trials. Inter-rater reliability was high; agreement between coders was 96% for coding proportion absolute error. To align the direction of scores for all early number skills assessments (with higher scores indicating better performance), we converted the proportion absolute error scores into proportion absolute accuracy (1 – proportion absolute error) for analyses.

Number Sets Identification (Geary et al., 2009). Children were shown a target number (5) and were asked to circle all of the "sets" that can be put together to equal the target value. A set was indicated by two or three linked boxes with objects inside of them. The sets included both Arabic numerals and concrete objects (see Figure 3). Children completed two pages and were given 60 seconds per page. The first page of the assessment consisted only of sets of concrete objects, while page two of the assessment consisted of sets that included both concrete objects and Arabic numerals. Instructions were not repeated before the child started the second page. For analyses, we used signal detection methods to calculate each child's sensitivity (d') in detecting the correct sets (Geary et al., 2009; MacMillan, 2002). The d' score is a standard score that takes into account the difference between the child's correct identifications and false alarms. Thus, children who circle many correct sets (e.g., "4 and 1" or "3 and 2") and few incorrect sets (e.g., "7 and 5" or "9 and 2") will have high sensitivity scores, whereas children who circle few correct sets and many incorrect sets will have low sensitivity scores. Inter-rater reliability was high; agreement between coders was 100% for coding total hits, total misses, total correct rejections, and total false alarms for each page.

Fast Counting (Schleifer & Landerl, 2011). Children were presented with a random number of dots ranging from 1 to 6 and were asked to say how many dots there were as quickly and accurately as they could. There were 24 trials (four for each number 1-6). Inter-rater reliability was high; agreement between coders was 100% for coding whether or not a given trial was correct. We used overall accuracy as the outcome variable in analyses.

Number Word Comprehension (Lipton & Spelke, 2005). Children were asked to map a target large number word (e.g., “sixty”) to a picture depicting the corresponding number of dots. On each trial, children were shown two cards. One card (the target) displayed the target number of dots (e.g., 60). The other card (the distractor) presented half as many dots or twice as many dots (e.g., 30 or 120). Children were asked to point to the card with the target number of shapes, and they were not given enough time to count. Thus, this task relied on children’s “natural” ability to quickly estimate and discriminate two sets of dots that differ by a 1:2 ratio, and then map the number word to the appropriate set. Target values tested were 20, 40, 60, 80, 100, and 120. Each participant received 24 test trials (12 with displays matched by dot size and 12 with displays matched by summed area) and eight control trials with a small-number array target, interspersed to maintain motivation and check that children were on task. After each response, children were given informative feedback. Inter-rater reliability was high; agreement between coders was 97% for coding the total number correct. We used overall accuracy as the outcome variable in analyses.

Mathematics Achievement. We assessed mathematics achievement with the widely used Woodcock-Johnson III achievement battery, specifically the Applied Problems subtest (WJ-III; Woodcock, McGrew, & Mather, 2001), which assesses the ability to analyze and solve practical mathematics problems. We chose to focus on this subtest because it is the most comprehensive and most appropriate for assessing young children’s mathematics achievement. It requires quantitative reasoning, problem solving, and mathematics knowledge. All problems are presented orally, and a visual stimulus of concrete objects, numbers, or text is provided as support. All analyses were conducted using Rasch-scaled W scores because we were interested in achievement growth over time. Average standard scores for Applied Problems at Time 1 (*M* = 103.58, *SD* = 14.98) were close to the normed average of 100, whereas standard scores at Time 2 were higher (*M* = 111.23, *SD* = 16.30). Inter-rater reliability was high; agreement between coders was 100% for coding the total number correct at both time points.

IQ Covariate. We assessed children's general cognitive ability using the Kaufman Brief Intelligence Test–Second Edition (KBIT–2) (Kaufman & Kaufman, 2004). Activities include matching vocabulary words to images, answering riddles, and identifying missing components in a pattern. Inter-rater reliability was high; agreement between coders was 100% for coding the IQ composite score (verbal and nonverbal components).

Procedure

Children completed the battery of measures reported here along with a battery of measures not included in this description as part of a large-scale longitudinal study designed to test how specific skills in kindergarten predict understanding of specific mathematics concepts (e.g., mathematical equivalence) in second and fourth grade. For the kindergarten assessments, children participated individually in two sessions, conducted two days apart. At the second grade assessment, children completed the WJ Applied Problems subtest. Sessions were conducted in a quiet room in a research lab or a local school. Each session lasted about 50 minutes. All tasks were presented to participants in a fixed order. The first session consisted of the KBIT-2, Quantity Discrimination, Number Sets Identification, and Dimensional Change Card Sort Task. The second session consisted of Fast Counting, Number Word Comprehension, Day/Night Stroop Task, Number Line Estimation, and WJ Applied Problems. In addition to parental consent, verbal assent was obtained from the child before each session. Children were also free to choose to stop participating at any time, and experimenters monitored children's non-verbal behavior to evaluate whether they may want to end the session.

Results

Analytic Approach

We tested a series of path models in Mplus v. 7.3 (Muthén & Muthén, 2012) to investigate children's specific early number skills as mediators of the association between executive functioning skills and mathematics achievement. Our first step was to test a multiple mediator model at the kindergarten time point with all five mediators entered simultaneously as a conservative test of indirect effects given that the common variance among the early number skills was accounted for in the model. We tested the significance of direct effects from executive functioning skills to each early number skill after controlling for age and IQ (Path a), the path from each early number skill to mathematics achievement after controlling for age and IQ (Path b), and the path from executive functioning skills to mathematics achievement after controlling for age and IQ (Path c'). In this model, we allowed the residuals of all specific early number skills to correlate. We examined indirect effects of the association between executive functioning skills and mathematics achievement through early number skills using the bias-corrected bootstrapping approach. This method of testing indirect effects is recommended by Preacher and Hayes (2008) (see also Shrout & Bolger, 2002) for multiple mediator models and does not assume normality of the indirect effect distribution. Using this method, we examined the unstandardized bootstrapped 95% confidence intervals (CI) to determine the significance of the indirect effects (significant effect is indicated by CIs that do not contain zero). We also tested an inverse model to rule out the possibility that children's executive functioning skills mediated the association between children's specific early number skills and mathematics achievement. These models were saturated and therefore had uninformative model fit statistics (i.e., perfect fit).

The kindergarten analysis revealed Number Sets Identification as a mediator between executive functioning skills and mathematics achievement. Thus, the next step was to test mediation in a longitudinal model to determine if Number Sets Identification continued to mediate the association between executive functioning skills and second grade mathematics achievement. In other words, we assessed indirect effects of the association between executive functioning skills and *growth* in mathematics achievement from kindergarten to second grade through Number Sets Identification. We tested the significance of direct effects from executive functioning skills to Number Sets Identification after controlling for age, IQ, and Time 1 mathematics achievement (Path a), the path from Number Sets Identification to Time 2 mathematics achievement after controlling for age, IQ, and Time 1 mathematics achievement (Path b), and the path from executive functioning skills to mathematics achievement after controlling for age, IQ, and Time 1 mathematics achievement (Path c'). We also regressed Time 1 mathematics achievement and executive functioning skills on age and IQ and allowed Time 1 mathematics achievement and executive functioning skills to correlate in the model. This model was also a saturated model and therefore had uninformative model fit statistics. It should be noted that the saturated model controls for kindergarten mathematics achievement in all paths, including the *a* path, even though it was the outcome of the cross-sectional model reported above. Our goal was to test the hypothesis that executive functioning skills predict Number Sets Identification, which then predicts longitudinal growth in math achievement (from kindergarten to second grade). The fully saturated model is the most conservative test of this hypothesized mediation effect because it accounts for prior mathematics achievement in all paths in the longitudinal model. Notably, results were the same when we did not control for kindergarten math achievement in the *a* path. Finally, we further explored significant indirect effects in follow-up analyses to further specify the skills that mediated the association between children's executive functioning skills and their mathematics achievement.

Descriptive Statistics and Correlations

Descriptive statistics are presented in Table 1 and zero-order correlations among all variables used in analyses are presented in Table 2. With the exception of the correlation between age and some of the mathematics variables, most variables were significantly correlated. Some of the strongest correlations emerged between Number Sets Identification and the WJ Applied Problems subtest at both time points and Number Sets Identification and Quantity Discrimination. The correlations between Number Word Comprehension and the other variables in the model were among the weakest correlations.

Two children opted not to complete Number Sets Identification and one child was not administered the advanced border version of the DCCS due to experimenter error. Additionally, 34 children who participated in kindergarten could not be relocated in second grade. When comparing scores from children who completed both time points of data to children with missing Time 2 data, we found that there were significant differences between these groups on Quantity Discrimination (t = 2.38, p = .019) and Number Sets Identification (t = 2.46, p = .015) and in each case, the group with missing data scored significantly lower compared to the group with complete data. Under the assumption that data is missing at random or missing not at random (which cannot be empirically verified), we followed the recommendations of Graham (2009) and used all available data for analyses using full information maximum likelihood (FIML) estimation.

Mediation Analyses

Prior to testing for indirect effects, we first examined the associations between executive functioning skills and mathematics achievement (as measured by the WJ Applied Problems subtest) at both Time 1 and Time 2, controlling for covariates. For analyses, each child's *Z* score for the Day/Night Stroop Task and *Z* score for the Dimensional Change Card Sort [DCCS] Task was averaged into an executive functioning composite *Z* score to be consistent with prior work suggesting that executive functioning is best represented as a unitary construct in young children (e.g., Wiebe et al., 2008). For the participant who was not administered the border version of the DCCS because of experimenter error, we used the Day/Night *Z* score instead of an average. Conclusions are unchanged if we impute a DCCS score for this participant based on a regression equation that predicts the DCCS score from other relevant variables. When we regressed Time 1 WJ Applied Problems onto IQ, age, and executive functioning skills, all three variables were predictive of Time 1 WJ Applied Problems: IQ ($\beta = .52$, $p < .001$), age ($\beta = .30$, $p < .001$), and executive functioning skills ($\beta = .21$, $p = .004$). We then regressed Time 2 WJ Applied Problems onto IQ, age, executive functioning skills, and Time 1 WJ Applied Problems. With the exception of age ($\beta = .03$, $p = .67$), the pattern of effects was similar. IQ, executive functioning skills and Time 1 WJ Applied Problems were predictive of Time 2 WJ Applied Problems: IQ ($\beta = .24$, $p = .002$), executive functioning skills ($\beta = .25$, $p < .001$), and Time 1 WJ Applied Problems ($\beta = .48$, $p < .001$).

Kindergarten Indirect Effects. We first tested our hypothesis that there would be a total indirect effect of the association between executive functioning skills and mathematics achievement through specific early number skills. In that model we also tested for significant specific indirect effects for each individual early number skill when accounting for other early number skills. The standardized path coefficients in Figure 1 include significant path coefficients for the association between kindergarten Number Sets Identification and Fast Counting and mathematics achievement, while children's executive functioning skills were associated with both Number Line Estimation and Number Sets Identification. Full model parameters can be found in Table 3. By examining the bootstrapped 95% CIs, we found that the total indirect effect of executive functioning skills on kindergarten mathematics achievement through early number skills was significant, and the specific indirect effect for Number Sets Identification was significant (see Table 4). The percentage of variance in kindergarten mathematics achievement explained by the model was 68.7%. Specific early number skills fully mediated the association between executive functioning skills and children's kindergarten mathematics achievement, as executive functioning skills were no longer a significant predictor of kindergarten mathematics achievement after adding specific early number skills to the model (c path - $\beta = .21$, $p = .004$; c' path - $\beta = .03$, $p = .63$). It is important to note that we also examined models in which Day/Night and the DCCS were treated separately rather than combined into a composite score and found the same pattern of effects. Because these data were assessed at a single time point, we tested for indirect effects of specific early number skills on kindergarten mathematics achievement through executive functioning skills to rule out the alternative specification of the model. We found that executive functioning skills did not mediate the associations between early number skills and mathematics achievement (all indirect effect bootstrapped 95% CIs contained 0).

Second Grade Indirect Effects. After establishing Number Sets Identification as a significant mediator of the association between executive functioning skills and mathematics achievement, we examined if that mediation effect held longitudinally. The indirect effect of executive functioning skills on gains in mathematics achievement from kindergarten to second grade through Number Sets Identification was significant, as the bootstrapped 95% CI did not include zero, estimate = 1.25, 95% CI [.29, 2.91] (see Figure 2 for standardized estimates of direct effects of interest and Table 5 for estimates of all direct effects). In contrast to the cross-sectional data, Number Sets Identification only partially mediated the relation between executive functioning skills and children's second grade mathematics achievement, as the path from executive functioning skills to children's second grade mathematics achievement was still significant after accounting for other paths in the model (c path - $\beta = .25$, $p = .004$; c' path - $\beta = .21$, $p = .01$). The percentage of variance in kindergarten mathematics achievement explained by the model was 70.2%. Again, we also examined models in which Day/Night and the DCCS were treated separately rather than combined into a composite score. In the longitudinal model, we found that the indirect effect from the DCCS to second grade mathematics achievement through Number Sets Identification was significant (estimate = .18, 95% CI [.01, 51]) whereas the model with Day/Night was not (estimate = .20, 95% CI [-.004, .61]), although the direct *a* and *b* paths coefficients were very similar and in the predicted direction. Note that we also ran a fully saturated longitudinal model that included all five specific early number skills as mediators. This model showed the same pattern of findings as the cross-sectional model in that only the indirect effect for Number Sets Identification had a bootstrapped 95% CI that did not include zero (see Appendix for all direct effects and indirect effects).

Follow-Up Analyses. Because Number Sets Identification was clearly the strongest mediator of the relation between executive functioning skills and mathematics achievement, and that effect held when examining gains in mathematics achievement, we probed the nature of the mediation effect for Number Sets Identification further in exploratory analyses.

First, we simply examined the types of errors children made on Number Sets Identification. Errors of omission, where a child did not circle a set that totaled five, were far more common than false alarms, where a child circled a set that did not total five (average error rates = 62.45% versus 9.39%). This difference in the error rates may not be surprising given that most children did not get to the last two rows on each page in the time allotted (only seven of 139 children finished all rows on page 1, and only eight of 139 children finished all rows on page 2). However, even excluding the last two rows from consideration in the error analysis, the average error rate for omissions (52.88%) was still significantly higher than the error rate for false alarms (10.32%), $t(138) = 16.77$, $p < .001$.

We reasoned that executive functioning skills would be particularly helpful for helping children focus on sets as wholes, rather than on their individual components, and that this ability to attend to sets as wholes while inhibiting individual salient components of a set would be helpful for learning the more difficult concepts and skills needed to succeed on a mathematics achievement test. Thus, we suspected that children's false alarms would be more likely to explain the overall indirect effect compared to omissions because false alarms represent children mistakenly circling a set that may have a salient distracting components (e.g., a five in the set; a set of shapes that is close in number to five) rather than inhibiting circling it. Omissions, in contrast, are more likely the result of insufficient early number skills alone.

To examine this prediction, we compared the indirect effects of children's executive functioning skills on their mathematics achievement through both false alarms and omissions. We tested path models for both kindergarten mathematics achievement as well as second grade mathematics achievement controlling for prior mathematics achievement. Specifically, we tested the significance of direct effects from executive functioning skills to false alarms and omissions after controlling for age and IQ (Path a), the paths from performance on both of these subscores of Number Sets Identification to mathematics achievement after controlling for age and IQ (Path b), and the path from executive functioning skills to mathematics achievement after controlling for age and IQ (Path c'). In the longitudinal models, we additionally controlled for kindergarten mathematics achievement as we did in previously discussed models. For the kindergarten model, the indirect effect through false alarms was significant (estimate = 2.35, 95% CI [.42, 5.07]), whereas the indirect effect through omissions was not (estimate = 1.92, 95% CI [-1.06, 5.08]). More importantly, when examining the direct paths in this model, the *a* path for false alarms regressed on executive functioning skills ($\beta = -.20$, $p = .03$) was significant, whereas the *a* path for omissions regressed on executive functioning skills was not ($\beta = -.13$, $p = .20$). In contrast, the direct *b* paths for mathematics achievement regressed on both variables were significant (false alarms $\beta = -.46$, $p < .001$; omission errors $\beta = -.61$, $p < .001$). The pattern of effects for second grade mathematics achievement was similar, although the indirect effect through false alarms did not reach significance (estimate = .98, 95% CI [-.05, 2.92]). As before, the indirect effect for omissions was not significant (estimate = .22%, 95% CI [-1.21, 1.88]). For the second grade model, the *a* path for false alarms regressed on executive functioning skills ($\beta = -.16$, $p = .09$) was marginal, while the *a* path for omissions regressed on executive functioning skills was not significant ($\beta = -.03$, $p = .76$). In contrast, the direct *b* paths for mathematics achievement regressed on both variables were significant (false alarms $\beta = -.23$, $p = .01$; omissions $\beta = -.29$, $p = .001$). These results suggest that stronger indirect effects were found through false alarms compared to omissions and that the differences between indirect effects for false alarms and omissions were more due to differences in how these different types of errors relate to executive functioning skills than to mathematics achievement.

We further speculated that because children were assessing whether sets totaled five, sets that did not total five but included a five (either symbolic or non-symbolic) might represent a particularly salient distractor for children. To examine this possibility, we first broke down the descriptive data for false alarms by sets that had a five as one of the components and sets that did not, and the percentage of false alarms committed on all sets that included a five was significantly higher (18.12%) than the percentage of false alarms committed on sets that did not include a five (6.24%), $t(138) = 11.78$, $p < .001$. This suggests that children were more likely to attend to a specific component of the set rather than focus on the set as a whole on trials that included a particularly salient component.

When we entered false alarms rates on sets that included a five and false alarms rates on sets that did not include a five together into a path model as indirect effects, neither by itself was a significant indirect effect in either the kindergarten or the second grade longitudinal model. This could be due to the collinearity between these variables, as they were highly correlated (r = .76). Therefore, we also explored false alarms on sets that included a five and sets that did not include a five in separate mediation models along with omission errors. For the kindergarten model including false alarms on sets with a five, we found that the indirect effect was significant (estimate = 1.31, 95% CI [.23, 3.26]) and that both the *a* path (β = -.22, p = .02) and *b* path were significant (β = -.24, p = .002). For the kindergarten model including false alarms on sets without a five, we also found that the indirect effect was significant (estimate = .07, 95% CI [.17, 4.42]) and that the *a* path was marginal (β = -.18, p = .06) while the *b* path was significant (β = -.40, p < .001). By comparing the standardized *a* paths, we can see that the *a* path for the false alarms on sets with a five was larger than the *a* path for the false alarms on sets without a five. For the second grade longitudinal model including false alarms on sets with a five, we found that while the indirect effect was not significant (estimate = .87, 95% CI [-.02, 2.79]), the *a* path was marginal (β = -.19, p = .06) and the *b* path was significant (β = -.18, p = .02). For the second grade longitudinal model including false alarms on sets without a five, we also found that the indirect effect was not significant (estimate = .76, 95% CI [-.14, 2.73]), but we found that the *a* path was not significant (β = -.13, p = .18) while the *b* path was significant (β = -.22, p = .04) was significant. By comparing the standardized *a* paths, we can see that the *a* path for the false alarms on sets with a five was larger than the a path for the false alarms on sets without a five. This suggests a pattern of effects that is consistent with the idea that sets that did not total five but included a five might represent a particularly salient distractor for children and that executive functioning skills would be especially helpful to attend to the whole set.

One might argue, however, that the false alarms on sets that included a five were simply the most difficult sets on the assessment, and that what we were capturing was close associations between children's ability to correctly respond to more difficult items generally rather than specifically their ability to inhibit attention to salient components of a number set. To explore this possibility, we first computed the percentage of children who got each set correct on the assessment. For sets that totaled five, we scored children as getting the set correct if they circled the set. For sets that did not total five, we scored children as getting the set correct if they did not circle the set. We then rank ordered item difficulty according to the percentage of children who got each set correct and computed a proportion correct score for each child for the 20 most difficult sets on the assessment. The first thing to point out is that of the 20 most difficult sets, only one of the sets was a false alarm on a set that included a five. Second, the most difficult sets were evenly distributed across page 1 and 2 of the assessment (page 1 contained only non-symbolic concrete quantities whereas page 2 contained both symbolic (Arabic numerals) and non-symbolic concrete quantities). Third, 19 of the 20 most difficult sets were sets that did add up to five, so children would have had to circle them to get them correct. We confirmed the non-overlap of false alarms on sets that included a five with the proportion correct on the 20 most difficult sets by computing a correlation between these two variables, $r = .03$, $p = .75$. Although not reported here, we also conducted additional mediation analyses comparing children's false alarm scores to their scores on the 20 most difficult items on the assessments as mediators in our models, and the patterns of results were consistent with the models reported above.

Discussion

Children were assessed on a battery of executive functioning skills tasks, early number skills tasks, an IQ test, and a nationally normed mathematics achievement measure to determine if specific early number skills mediate the association between children's executive functioning skills and growth in mathematics achievement from kindergarten to second grade. We first established that executive functioning skills were significantly associated with mathematics achievement at both kindergarten and second grade in our sample after controlling for covariates. Next, we entered the specific early number skills together into a multiple mediator model to determine if the combined effect of these skills mediated the association between executive functioning skills and mathematics achievement. This yielded an overall significant indirect effect of the association between executive functioning skills and kindergarten mathematics achievement through specific early number skills. This analysis indicated full mediation such that executive functioning skills were no longer associated with kindergarten mathematics achievement once specific early number skills were entered into the model. Within the set of early number skills mediators, we found that Number Sets Identification was driving the significant overall indirect effect. These results did not differ from alternative models in which each executive functioning measure was entered separately as a predictor rather than using a composite score. We also tested a reverse mediation pathway in which executive functioning skills mediated the associations between specific early number skills and mathematics achievement and were not able to support it as an alternative explanation for the associations.

We then tested if these findings held longitudinally. We found a significant indirect effect of the association between executive functioning skills and children's mathematics achievement in second grade through Number Sets Identification after controlling for kindergarten mathematics achievement. Unlike the kindergarten analysis, however, we found partial mediation such that executive functioning skills remained a significant predictor of second grade mathematics achievement even after entering Number Sets Identification into the model. When we examined each executive functioning skills measure separately rather than as a composite, we found that the mediation effect was slightly stronger for the model using the DCCS as a single indicator of executive functioning skills compared to a model with Day/Night as the indicator, although the path coefficients were similar. Upon further exploration of Number Sets Identification, we found more specifically that the ability to inhibit attention to salient components of a set in favor of attending to a number set as a whole accounted, at least in part, for the significant indirect effects.

Based on these results, we conclude that one possible mechanism through which executive functioning skills are associated with mathematics achievement is through early number skills, specifically attention to number sets. Children's ability to attend to a number set as a whole rather than focus on specific components of that set requires executive functioning skills and may be particularly important for mathematics achievement in elementary school. In older middle-school aged children, Siegler and Pyke (2013) found support for a specific link between executive functioning skills and fraction knowledge, suggesting that executive functioning skills may have allowed children to attend to the fraction as a whole rather than specific components of the fraction. In the current study, we saw a similar effect in younger children. The data pointed to false alarms in Number Sets Identification as the driver of the association between executive functioning skills and children's mathematics achievement, and omissions seemed to be less relevant. We also found stronger associations between false alarms on sets that included a five and executive functioning skills compared to false alarms on sets that did not include five. We reasoned that false alarms on sets that included a five might best represent children's tendency to focus on one salient component of a set rather than examining the set as a whole.

An alternative explanation of the current findings is that Number Sets Identification was simply the best assessment of early mathematics skills. In other words, critics may say that it is not the ability to attend to number sets *per se* that explains the link between executive functioning skills and mathematics achievement, but rather that Number Sets Identification is simply the best indicator of early mathematics achievement. However, we found that there was an indirect effect of the association between executive functioning skills and second grade mathematics achievement through Number Sets Identification even when we controlled for kindergarten mathematics achievement. In fact, we included kindergarten mathematics achievement as a covariate in all paths in the model. Thus, the second grade mediation model accounts for the overlap between Number Sets Identification and general mathematics achievement and Number Sets Identification remained a significant mediator of the association between executive functioning skills and children's later mathematics achievement. This suggests that Number Sets Identification was not simply assessing mathematics achievement but was assessing a unique specific early number skill.

We cannot rule out the possibility, however, that there may be other unmeasured specific early number skills that may also be mechanisms explaining the link between executive functioning skills and mathematics achievement gains. We also cannot rule out the possibility that we are overestimating the effects of executive functioning skills and early number skills on future mathematics achievement. As Bailey (2014) cogently argues, correlational studies—even when they are prospective studies that control for important stable characteristics—almost always overestimate the effects of early skills on future achievement. This is because links between knowledge of mathematical concepts across time may be an index of unmeasured stable traits related to math learning, such as motivation to learn, attention to detail, or abstract reasoning skills (Bailey, Watts, Littlefield, & Geary, 2014). Although our analyses control for IQ, which is likely to be highly correlated with these stable unmeasured traits, it was not possible for us to statistically control for all possible stable characteristics that may be associated with mathematics achievement (Bailey, 2014).

Overall, our findings do not imply that other specific early number skills are not important predictors of mathematics achievement, nor do they suggest that these skills are not influenced at least in part by executive functioning skills. Rather, these skills may not explain the association between executive functioning skills and mathematics achievement. Number Sets Identification was more strongly associated with executive functioning skills compared to the other specific early number skills measures, which could lead one to suspect that perhaps Number Sets Identification and executive functioning skills are tapping into the same underlying construct and that is why we found Number Sets Identification to be a mediator while the other specific early number skills were not mediators. However, taking a close look at the correlations in Table 2, it is clear that the correlations between Number Sets Identification and the executive functioning measures are very similar to the correlations between Number Line Estimation and executive functioning skills. Indeed, research has shown links between executive functioning skills, number line estimation, and mathematics achievement, suggesting the possibility that Number Line Estimation could have been a mediator (Geary et al., 2008; Sasanguie et al., 2012). Number Line Estimation requires children to attend to the number line as a whole instead of being drawn to the 0 endpoint and relying on a typical left-to-right processing strategy. If the mediation effect was truly due only to the overlapping executive functioning requirements that involve inhibiting salient information and strategies to produce a correct response and not to the ability to attend to a number set rather than becoming distracted by components of that set, then we would expect for both number line estimation and number sets to be mediators, which we did not find. That said, it is possible that Number Line Estimation does not involve the same overlapping EF requirements, or that another procedure for administering Number Line Estimation may have yielded mediation effects. We used the procedure in which children are presented with a number line (e.g., 0-20) and then are given a familiarization trial in which 0, 20, and 10 are all placed on the line to ensure that children understand the task (as in Barth & Paladino, 2011 and Booth & Siegler, 2006). However, this “anchored” administration leads to more linear estimates, which may obscure variability and lessen the chances of finding an effect of Number Line Estimation on subsequent mathematics achievement (see Opfer, Thompson, & Kim, 2016). Showing the placement of 10 at the midpoint also may have helped children inhibit their natural tendency to start at 0 use a left-to-right processing strategy

Based on the idea that attention to sets as a whole is an important mediator of the association between executive functioning skills and mathematics achievement, one might have expected Number Word Comprehension to be a significant mediator. This task requires children to match verbally presented symbolic number words to the correct non-symbolic quantity out of two choices, some trials of which have conflicting surface area information. Based on an account that specific early number skills that require attention to a number set when salient or distracting components of the set are present, we would expect that children's accuracy on this task could potentially require executive functioning skills to inhibit either salient symbolic information or distracting surface area cues. One potential explanation for this null finding is that the Number Word Comprehension task may have been too difficult for our sample and not truly representative of individual differences in children's skills. In fact, for "larger" (80/100/120) targets, children's average accuracy was about 50% (chance). However, even when we examined a mediation model with only "smaller" (20/40/60) targets, children's accuracy on Number Word Comprehension remained a non-significant mediator. It may be that this task primarily assesses understanding of number word vocabulary.

One related early number skill that also requires attention to a number set when salient or distracting components of the set are present is approximate number system (ANS) acuity. ANS acuity refers to the precision with which individuals can compare and discriminate large numbers of non-symbolic items (e.g., dots) without counting (Feigenson, Dehaene, & Spelke, 2004). It has been shown to be associated with both inhibitory control (e.g., Clayton & Gilmore, 2015; Fuhs & McNeil, 2013) and mathematics achievement (e.g., Halberda, Mazzocco, & Feigenson, 2008; Libertus, Feigenson, & Halberda, 2011) in past work, so it might seem like a good candidate for a mediator in the present study. However, we chose not to include a typical ANS acuity task because of questions surrounding the measurement of ANS acuity (Clayton & Gilmore, 2015; Dietrich, Huber, & Nuerk, 2015). Moreover, recent studies have presented evidence that the association between ANS acuity and mathematics achievement is not significant when other specific early number skills (e.g., symbolic magnitude comparison - Lyons, Price, Vaessen, Blomert, & Ansari, 2014) and executive functioning skills—specifically inhibitory control (Clayton & Gilmore, 2015; Fuhs & McNeil, 2013; Gilmore et al., 2013) are taken into account. It is important to note that this latter finding has not been interpreted as an indirect effect of the association between ANS acuity and mathematics achievement through inhibitory control, but rather as a confounding effect of inhibitory control on the association between ANS acuity and mathematics achievement. Indeed, there is substantial overlap in some of the tasks used to measure ANS acuity and inhibitory control (e.g., Merkley, Thompson, & Scerif, 2016). We acknowledge that it is possible that we missed a potentially important early number skill, and we cannot rule out the possibility that we could have found evidence for the reverse mediation pathway in which executive functioning skills mediate the association between early number skills and mathematics achievement if we had included ANS acuity in our set of early number skills.

Overall, the present findings support a growing empirical literature showing significant direct associations between children's executive functioning skills and their specific early number skills (e.g., Geary et al., 2007; Kroesbergen et al., 2009), and the second grade model supports previous findings of significant direct associations between children's executive functioning skills and their longer-term mathematics achievement growth (e.g., Fuhs et al., 2014). The findings extend prior work in important ways, however, because they suggest a specific way in which these prior findings can be integrated theoretically into a longitudinal account of associations between domain-general executive functioning skills and domain-specific mathematics achievement.

Two differences between the findings from the kindergarten and second grade models are worth noting. The first is that we found partial mediation in the second grade model but full mediation in the kindergarten model. Even when we ran both models with the full set of specific early number skills as mediators, executive functioning skills continued to predict second grade mathematics achievement after including the mediators and controlling for kindergarten mathematics achievement. This could be explained by the fact that both the DCCS and Day/Night were both more strongly correlated with second grade mathematics achievement compared to kindergarten mathematics achievement (see Table 2). The items administered to children on the Applied Problems subtest of the WJ-III, our assessment of mathematics achievement, are different across time as they are based on children's age and performance. It could be that the more complex problem solving items that appear later in the assessment actually rely more heavily on executive functioning skills compared to earlier items on the assessment, although both scores at kindergarten and second grade were significantly related to executive functioning skills.

The other difference that emerged when comparing the findings of the kindergarten and second grade models was that when the executive functioning skills measures were examined in separate models (rather than a composite), only the indirect effect for the association between the DCCS and second grade mathematics achievement through Number Sets Identification achieved significance (the indirect effect that included only Day/Night in the model did not). However, the path coefficients were similar, and in the kindergarten model, the results for both the models with the DCCS and Day/Night entered individually were similar to the results of the model with the composite measure. A composite measure is widely considered to be the best representation of executive functioning skills in young children. This is not only because there is support for a one-factor model of executive functioning skills in young children (e.g., Weibe et al., 2008), but also because executive functioning skills assessments suffer from the task impurity problem (Miyake et al., 2000). Specifically, it is difficult to create a task that is purely working memory, inhibitory control, and/or cognitive flexibility while not also tapping the other skills, and executive functioning skills assessments also likely have considerable random noise due to the fact that executive functioning skills assessment also tap young children's language skills, motor skills, or other skill required for the particular task. Despite these caveats, there is some evidence for empirical differentiation of executive functioning skills performance over time (Lee et al., 2008), which could suggest that individual measures of executive functioning skills may differentially predict mathematics achievement at later time points in development. This is consistent with our finding that DCCS seemed to predict second grade achievement better than Day/Night. The DCCS is regarded as a more complex measure of executive functioning skills, as it requires cognitive flexibility as well as response inhibition, and cognitive flexibility tends to develop somewhat later than (and thus builds on) other executive functioning skills (Garon, Bryson, & Smith, 2008). For this reason, DCCS may be a more robust predictor of children's mathematics achievement in second grade as children learn to solve more complex mathematical problems. It is also a task that requires children to form a "mental set" (i.e., representation) of which dimensions to attend to and which to ignore in the pre-switch phase (Garon et al., 2008) that is difficult to modify later. That we found the DCCS model to indicate indirect effects in second grade could strengthen the argument that it is specifically the ability to flexibly attend to a “mental set,” or “rule” underlying the mediation effect, which in the case of Number Sets Identification would be the target numerosity of a set totaling five.

The present results may have practical implications for educational environments for elementary school children, as they suggest several possible targets for intervention to improve children’s mathematics achievement: executive functioning skills, attention to number sets, or both. Whether executive functioning skills are malleable and whether intervention designed to improve executive functioning skills would lead to improvements in attention to number sets as wholes are currently open questions that require future research. Working memory training studies indicate that effects are variable across studies, with mostly short-term, specific training effects that do not generalize (Melby-Lervåg & Hulme, 2013; Thorell, Lindqvist, Bergman Nutley, Bohlin, & Klingberg, 2009; see also Redick, 2015). Longer-term field-based work has suggested that executive functioning skills may be improved through interventions both directly (Tominey & McClelland, 2011) and indirectly (Raver et al., 2011), but other curricular interventions that are designed to target executive functioning skills (*Tools of the Mind*, Bodrova & Leong, 2007) have shown mixed results (e.g., Blair & Raver, 2014; Farran & Wilson, under review). Further, to date, there are no studies in which researchers have attempted to train executive functioning skills to see if it improves children’s attention to number sets. Researchers have looked at whether training in executive functioning skills improves children’s mathematics achievement, with two studies finding effects for English language learners (Blair & Raver, 2014; Schmitt, McClelland, Tominey, & Acock, 2015), and others finding no effects (e.g., Farran & Wilson, under review). Clearly, more work is needed to determine if executive functioning skills are malleable and if improvements in executive functioning skills would lead to improvements in attention to number sets and mathematics achievement more generally.

Given the limits of the research on executive functioning skills interventions, a more promising route for training to improve children's mathematics achievement may be training specifically on attention to number sets. Recent work has pointed to several early number interventions that have shown both direct effects on the early number skills targeted and promise for promoting later mathematics achievement (e.g., Clements & Sarama, 2007; Jordan et al., 2013; Ramani & Siegler, 2008). Our results show that attention to number sets may be important for mathematics achievement and mediates the link between executive functioning skills and children's growth in mathematics achievement; thus, an increasing emphasis on training specific mathematics skills that require children to attend to a number set as a whole rather than to components of that set may be warranted as a possible means to improving children's mathematics achievement. This ability to attend to a number set as a whole may be an early skill that is so central to the learning of mathematics that it makes it easier for children to learn more advanced mathematics concepts in school (cf. Charles, 2005). It may be an earlier form of the "structure sense" described by mathematics education researchers that allows children to see a number set as an abstract, manipulable object in its own right, which can aid in flexible problem solving and creative manipulation of numbers (Linchevski & Livneh, 1999; Sfard, 1991). For example, if children can treat five fingers as a whole set with the sum of five, then they will not need to count out "one, two, three, four, five" every time they are trying to combine five with another number (i.e., counting all versus counting on, Siegler, 1987, 1999). Similarly, if children can treat $3 + 2$ as an object (with a sum of five), and also understand $2 + 3$ as an object with a sum of five, then, through transitive reasoning processes, children can come to understand the higher-level concept that $3 + 2 = 2 + 3$ (the commutative property). In support of this view, McNeil and colleagues (2012) found that experimentally manipulating the organization of addition fact practice into groups by equivalent sums (e.g., 1 + 4 = __, 2 + 3 = __) leads to an improved formal understanding of mathematical equivalence, the relation between two quantities that are equal and interchangeable (see also Kieran, 1981).

The scenario in which salient non-symbolic and symbolic information is placed together during mathematics activities where children have to attend to sets is not uncommon in early educational activities (e.g., matching pictures sets [e.g., five ducks] with their corresponding symbolic numerals [e.g., 3, 5, 7], using manipulatives when learning to add small sets together, or playing games like War or Go Fish with a standard deck of cards). Moreover, the ability to attend to a number set as a whole in the face of other distracting information seems particularly important for learning early mathematics concepts like cardinality, the order-irrelevance principle in counting, the "count on," "min," and decomposition strategies for solving addition problems, and the commutative property of addition. Recent research has supported connections among these concepts, especially in children with a mathematics learning disability (e.g., Geary, Hoard, Byrd-Craven, & DeSoto, 2004; Geary et al., 2007) and highlight the importance for future research to focus on number sets identification. Specifically, Geary and colleagues (2007) found that number sets identification predicted individual differences in children's use of the "min" strategy for solving complex addition problems. Additionally, numerosity coding (Butterworth, 2010; i.e., the understanding of cardinal value and the notion that smaller-magnitude numbers make up larger numbers) is a key skill involved in number sets identification. Beyond number recognition and naming, skill at both composing and decomposing quantities is critical for success at number sets identification, and likely related to the use of decomposition in solving arithmetic problems (Geary, 2011) and understanding of mathematical equivalence (Chesney et al., 2014; Hornburg, McNeil, Devlin, & McKeever, 2015).

That there is more experimental evidence currently for the malleability of specific early number skills to improve mathematics achievement than there is for the malleability of executive functioning skills to improve children's attention to number sets does not necessarily rule out the possibility that a combined intervention would yield significant benefits to children. However, empirical support for a combined executive functioning skills and specific early number skills intervention is limited (e.g., Barnes et al., 2014). If a combined executive functioning skills and early number skills curriculum offers specific benefits above and beyond either curriculum alone, then we would expect the combined curriculum to produce greater gains in both executive functioning skills and specific number skills in pre-kindergarten children. However, Clements and colleagues (Clements, Sarama, Unlu, & Layzer, 2012) found that a combined intervention did not produce greater benefits to children's executive functioning skills compared to the *Building Blocks* curriculum, and in fact there was some evidence that the *Building Blocks* curriculum alone may have benefited children's executive functioning skills. This could mean that specific early number skills interventions produce more gains for children in *both* specific early number skills and executive functioning skills, or it could mean that the particular executive functioning skills intervention used was not an ideal intervention to enhance children's executive functioning skills.

Limitations

At least three specific limitations to this study must be highlighted. First, although we assessed children on a range of early number skills that have generally been shown to be associated with executive functioning skills, we did not include all possible skills that could promote mathematics achievement. Thus, we may not have included all potential mediators of the association between executive functioning skills and mathematics achievement. It is possible that there are more global individual and contextual factors at work that affect our variables of interests similarly. However, we did use a fairly strict set of controls, including prior performance on mathematics as well as IQ and age, and the fact that we found Number Sets Identification to be a significant mediator, both cross-sectionally and longitudinally, even after controlling for IQ, offers important insight into the types of early number skills that are likely candidates as mediators.

Second, we had information on children’s mathematics achievement at two time points, but we did not have multiple time points of measurement for all variables in the model. A longitudinal mediation model that fully accounts for growth in the mediators as well as the outcomes will be an important future study to consider.

Third, we did not have enough measures of each component skill of executive functioning skills to be able to utilize a latent variable approach to capturing executive functioning skills. We, therefore, cannot assess the dimensionality of the construct or give specific indications about potential differences among components skills in their relationship with mathematics achievement and specific early number skills. However, we did use two different measures of executive functioning skills that have been widely used and validated in the literature. Also, the patterns of findings remained largely unchanged regardless of which measure of executive functioning skills we used when we included them as individual predictors in our models rather than as a composite.

Conclusions

We found evidence that specific early number skills, and Number Sets Identification in particular, mediate the associations between children's executive functioning skills and mathematics achievement growth. These results suggest that one possible mechanism explaining the link between children's executive functioning skills and mathematics skills development is the ability to attend to a number set as a whole, rather than to the salient components of the set. Our findings highlight future directions for research, including research aimed at gaining a better understanding of children's ability to attend to number sets and developing targeted instruction on attention to number sets to test if improvements in number sets identification skills lead to improvements in mathematics achievement.

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Table 1

Descriptive Statistics

	N	Min.	Max.	*M*	*SD*	Skewness	Kurtosis
Kindergarten Age	141	5.32	7.38	6.16	0.36	0.46	0.49
IQ	141	56.00	132.00	100.04	13.69	-0.41	0.82
DCCS (total correct on post-switch trials)	141	0.00	18.00	12.76	4.38	-1.48	2.56
Day/Night (total correct)	140	1.00	16.00	12.74	3.43	-1.37	1.69
Quantity Discrimination (total correct)	141	0.00	42.00	25.68	7.27	-0.67	1.97
Number Line Estimation (accuracy)	141	0.50	0.98	0.86	0.10	-1.49	1.54
Number Sets Identification (d' score)	139	-3.20	3.78	0.00	1.32	0.26	0.05
Fast Counting (accuracy)	141	0.04	1.00	0.90	0.13	-2.83	13.51
Number Word Comprehension (accuracy)	141	0.41	0.88	0.67	0.10	-0.12	0.00
Kindergarten WJ Applied Problems (W score)	141	393.00	507.00	439.23	20.62	0.28	0.34
Second Grade WJ Applied Problems (W score)	107	423.00	542.00	489.79	21.96	-0.68	0.81

Note. Children's d' scores (sensitivity scores) on Number Sets Identification were calculated from the difference between a child's hit rate *Z* score and false alarm rate *Z* score. Thus, d' scores are standardized scores.

Table 2

Correlations

	1	2	3	4	5	6	7	8	9	10
1. Kindergarten Age	1									
2. IQ	-.185*	1								
3. DCCS	.165	.503**	1							
4. Day/Night	.111	.278**	.316**	1						
5. Quantity Discrimination	.263**	.442**	.401**	.198*	1					
6. Number Line Estimation	.145	.433**	.462**	.315**	.602**	1				
7. Number Sets Identification	.254**	.524**	.511**	.328**	.587**	.470**	1			
8. Fast Counting	.062	.352**	.368**	.164	.316**	.390**	.448**	1		
9. Number Word Comprehension	.091	.232**	.204*	.100	.336**	.278**	.335**	.167*	1	
10. Kindergarten Mathematics Achievement	.239**	.571**	.534**	.307**	.570**	.524**	.778**	.470**	.383**	1
11. Second Grade Mathematics Achievement	.121	.613**	.649**	.370**	.575**	.537**	.737**	.445**	.445**	.755**

*$p < .05$. **$p < .01$.

Table 3

Direct Effects: Predictors of Kindergarten Mathematics Achievement

Path	β	Estimate	95% CI
Kindergarten Mathematics Achievement ON			
Age	0.12	7.00	1.33, 14.19
IQ	0.20	0.31	0.11, 0.56
Executive Functioning Skills	0.03	0.69	-2.34, 3.16
Quantity Discrimination	0.03	0.07	-0.32, 0.42
Number Line Estimation	0.07	14.23	-13.34, 42.88
Number Sets Identification	0.49	7.46	5.09, 10.26
Fast Counting	0.12	19.54	2.71, 40.27
Number Word Comprehension	0.09	18.78	-2.35, 42.44
Quantity Discrimination ON			
Executive Functioning Skills	0.07	0.62	-1.01, 2.26
Age	0.34	6.85	3.97, 10.23
IQ	0.47	0.25	0.16, 0.37
Number Line Estimation ON			
Executive Functioning Skills	0.27	0.04	0.01, 0.06
Age	0.16	0.05	0.01, 0.09
IQ	0.33	0.002	.001, .004
Number Sets Identification ON			
Executive Functioning Skills	0.27	0.45	0.23, 0.66
Age	0.28	1.07	0.61, 1.54
IQ	0.48	0.05	0.03, 0.06
Fast Counting ON			
Executive Functioning Skills	0.16	0.03	-0.01, 0.08
Age	0.09	0.03	-0.06, 0.10
IQ	0.29	0.003	0.001, 0.005
Number Word Comprehension ON			
Executive Functioning Skills	0.05	0.01	-0.02, 0.03
Age	0.13	0.03	-.004, 0.08
IQ	0.23	0.002	0.001, 0.003
Executive Functioning Skills ON			
Age	0.27	0.60	0.29, 0.93

IQ	0.53	0.03	0.02, 0.04
Quantity Discrimination WITH			
Number Line Estimation	0.46	0.24	0.14, 0.37
Number Sets Identification	0.40	2.31	1.34, 3.40
Fast Counting	0.14	0.10	-0.03, 0.27
Number Word Comprehension	0.23	0.13	0.04, 0.25
Number Line Estimation WITH			
Number Sets Identification	0.22	0.02	0.01, 0.03
Fast Counting	0.23	0.002	0.00, 0.01
Number Word Comprehension	0.17	0.001	0.00, 0.003
Number Sets Identification WITH			
Fast Counting	0.24	0.03	0.01, 0.05
Number Word Comprehension	0.27	0.02	0.01, 0.04
Fast Counting WITH			
Number Word Comprehension	0.07	0.001	-0.001, 0.003

Table 4

Total Indirect Effects and Specific Indirect Effects: Executive Functioning Skills and Kindergarten Mathematics Achievement through Specific Early Number Skills

	Estimate	95% CI
Sum of Indirect Effects	4.52	2.04, 7.49
Quantity Discrimination	0.05	-0.20, 0.80
Number Line Estimation	0.50	-0.39, 1.76
Number Sets Identification	3.38	1.63, 6.05
Fast Counting	0.49	-0.14, 1.72
Number Word Comprehension	0.11	-0.24, 1.14

Table 5

Direct Effects: Predictors of Second Grade Mathematics Achievement

Path	β	Estimate	95% CI
Second Grade Mathematics Achievement ON			
Age	-0.01	-0.36	-7.78, 7.47
IQ	0.19	0.31	0.08, 0.54
Executive Functioning Skills	0.21	5.60	1.80, 9.97
Number Sets Identification	0.32	5.17	1.80, 8.99
Kindergarten Mathematics Achievement	0.28	0.30	0.09, 0.51
Number Sets Identification ON			
Executive Functioning Skills	0.15	0.24	0.06, 0.42
Age	0.11	0.40	-0.02, 0.80
IQ	0.16	0.02	0.003, 0.03
Kindergarten Mathematics Achievement	0.59	0.04	0.03, 0.05
Kindergarten Mathematics Achievement ON			
Age	0.36	20.49	13.47, 27.60
IQ	0.64	0.96	0.76, 1.18
Executive Functioning Skills ON			
Age	0.27	0.60	0.29, 0.93
IQ	0.53	0.03	0.02, 0.04
Kindergarten Mathematics Achievement WITH			
Executive Functioning Skills	0.23	2.39	0.69, 4.49

Figure Captions

Figure 1. Indirect effects of the association between executive functioning skills and kindergarten mathematics achievement through specific early number skills.

Note: Age and IQ were also covaried out of all paths in the analysis. Mediators were allowed to correlate in the model but correlations are not depicted here for parsimony. Standardized parameter estimates are reported. *$p < .05$. **$p < .01$.

Figure 2. Indirect effect of the association between executive functioning skills and growth in mathematics achievement from kindergarten to second grade through number sets identification.

Note: Age and IQ were also covaried out of all paths in the analysis. Standardized parameter estimates are reported. *$p < .05$. **$p < .01$.

Figure 3. Examples of the types of sets included on Number Sets Identification.

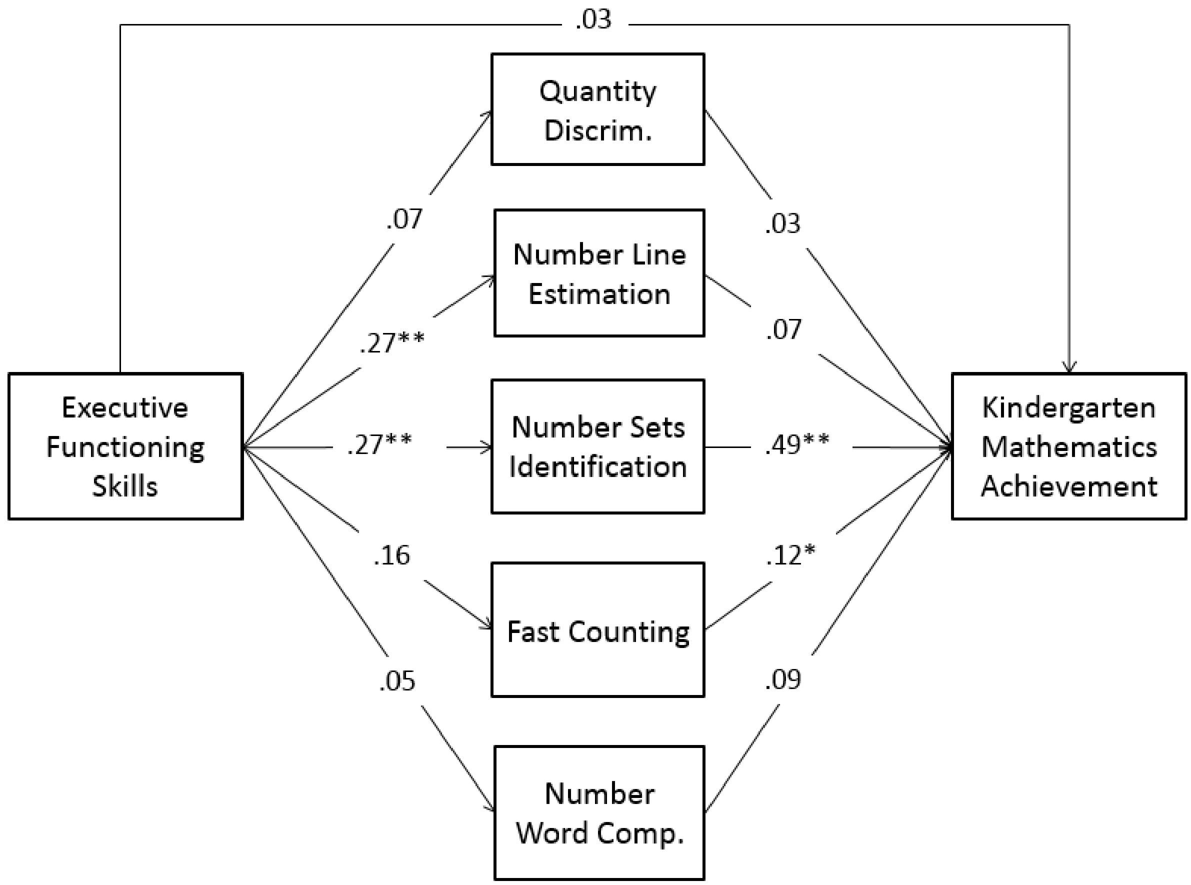

Figure 1 (see previous page for caption).

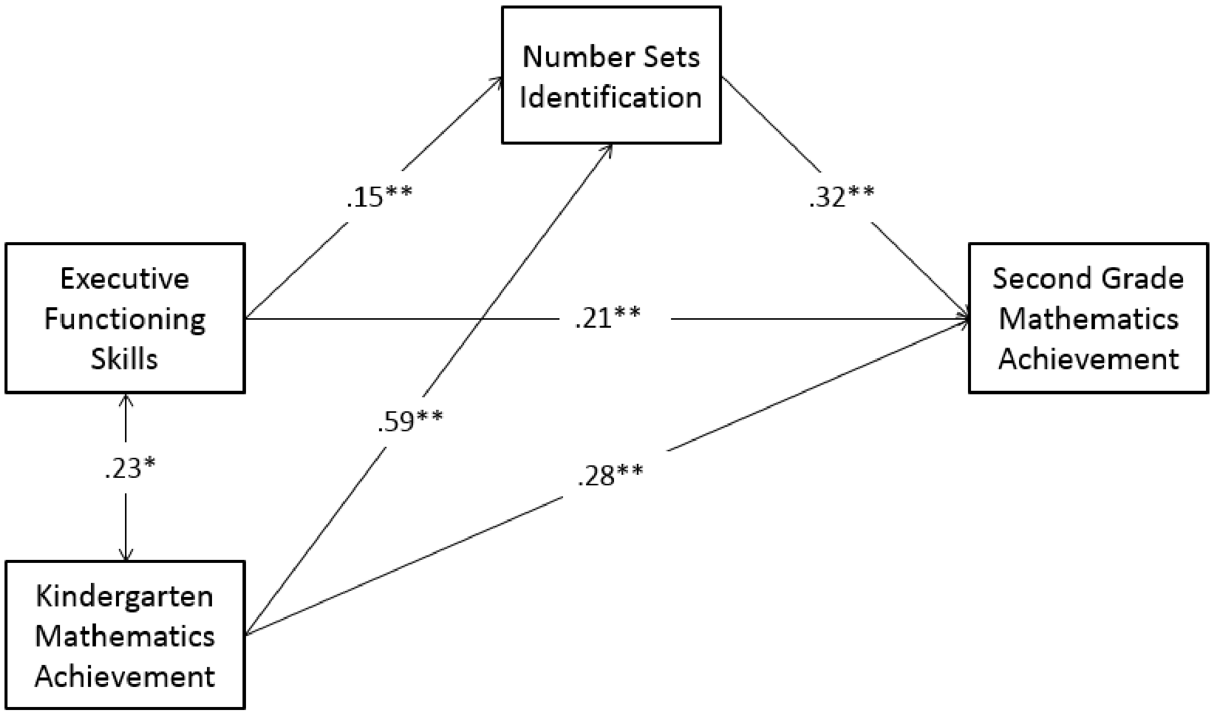

Figure 2 (see previous page for caption).

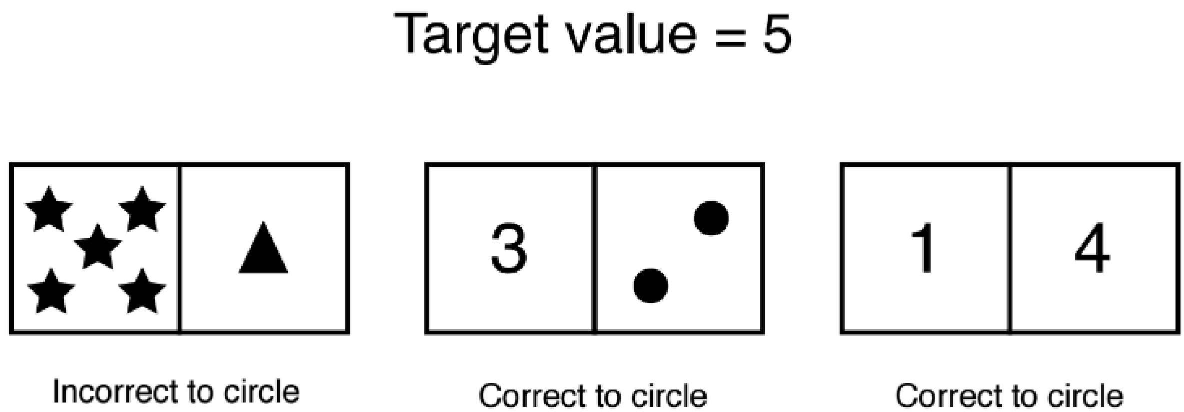

Figure 3 (see previous page for caption).

Appendix

Direct Effects: Including all Specific Early Number Skills as Predictors of Second Grade Mathematics Achievement

Path	β	Estimate	95% CI
Second Grade Mathematics Achievement ON			
Age	-0.01	-0.01	-0.08, 0.07
IQ	0.17	0.003	0.001, 0.01
Kindergarten Mathematics Achievement	0.20	0.21	-0.01, 0.41
Executive Functioning Skills	0.21	0.06	0.02, 0.10
Quantity Discrimination	0.13	0.004	-0.001, 0.01
Number Line Estimation	-0.02	-0.05	-0.37, 0.26
Number Sets Identification	0.24	0.04	0.01, 0.08
Fast Counting	0.09	0.15	-0.15, 0.33
Number Word Comprehension	0.17	0.39	0.13, 0.61
Quantity Discrimination ON			
Age	0.23	4.68	1.70, 7.67
IQ	0.28	0.15	0.06, 0.27
Executive Functioning Skills	-0.003	-0.03	-1.69, 1.78
Kindergarten Mathematics Achievement	0.35	12.5	5.47, 18.60
Number Line Estimation ON			
Age	0.07	0.02	-0.02, 0.06
IQ	0.17	0.001	0.00, 0.003
Executive Functioning Skills	0.21	0.03	0.003, 0.05
Kindergarten Mathematics Achievement	0.30	0.15	0.02, 0.27
Number Sets Identification ON			
Age	0.10	0.39	-0.04, 0.80
IQ	0.17	0.02	0.004, 0.03
Executive Functioning Skills	0.15	0.26	0.07, 0.45
Kindergarten Mathematics Achievement	0.59	3.87	2.95, 4.63
Fast Counting ON			
Age	-0.03	-0.01	-0.10, 0.06
IQ	0.09	0.001	-0.001, 0.002
Executive Functioning Skills	0.08	0.01	-0.02, 0.06
Kindergarten Mathematics Achievement	0.39	0.24	0.12, 0.39

Number Word Comprehension ON			
Age	0.01	0.003	-0.04, 0.05
IQ	0.03	0.00	-0.001, 0.002
Executive Functioning Skills	-0.03	-0.003	-0.03, 0.02
Kindergarten Mathematics Achievement	0.38	0.24	0.06, 0.28
Executive Functioning Skills ON			
Age	0.27	0.60	0.29, 0.93
IQ	0.53	0.03	0.02, 0.04
Kindergarten Mathematics Achievement ON			
Age	0.36	0.21	0.14, 0.28
IQ	0.64	0.01	0.01, 0.01
Executive Functioning Skills WITH			
Kindergarten Mathematics Achievement	0.23	0.02	0.01, 0.05
Quantity Discrimination WITH			
Number Line Estimation	0.41	0.19	0.12, 0.31
Number Sets Identification	0.29	1.28	0.60, 2.22
Fast Counting	0.05	0.03	-0.08, 0.15
Number Word Comprehension	0.16	0.08	0.00, 0.20
Number Line Estimation WITH			
Number Sets Identification	0.09	0.01	-0.01, 0.02
Fast Counting	0.17	0.002	0.00, 0.004
Number Word Comprehension	0.10	0.001	-0.001, 0.002
Number Sets Identification WITH			
Fast Counting	0.07	0.01	-0.01, 0.02
Number Word Comprehension	0.14	0.01	0.00, 0.02
Fast Counting WITH			
Number Word Comprehension	-0.02	0.00	-0.002, 0.002

Note: Because the scores on mathematics achievement (WJ Applied Problems) were on a much larger scale than the scores for other measures in the model, we transformed the scores by dividing them by 100 to align the scale of the measures to improve convergence.

Total Indirect Effects and Specific Indirect Effects: Executive Functioning Skills and Second Grade Mathematics Achievement through Specific Early Number Skills

	Estimate	95% CI
Sum of Indirect Effects	0.01	-0.01, 0.03
Quantity Discrimination	0.00	-0.01, 0.01
Number Line Estimation	-0.001	-0.02, 0.01
Number Sets Identification	0.01	0.001, 0.03
Fast Counting	0.002	-0.002, 0.02
Number Word Comprehension	-0.001	-0.01, 0.01

Note: Because the scores on mathematics achievement (WJ Applied Problems) were on a much larger scale than the scores for other measures in the model, we transformed the scores by dividing them by 100 to align the scale of the measures to improve convergence.